CMUC Online Seminar on Algebra, Logic and Topology

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Actions of preordered monoids and their associated extensions

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Abstract

The category of preordered monoids can be seen either as a category of internal preorders in monoids or as the category of internal monoids in preorders. In this talk, we start by analyzing differences and similarities between preordered groups and preordered monoids. In particular, we study the behavior of the forgetful functors of the latter category into the category of monoids and into the one of preorders. Then we restrict our attention to the category of preordered monoids whose preorder is determined by its cone of positives and prove that it is isomorphic to the category of right normal monoid monomorphisms. Finally, we describe Schreier split extensions in the category of right normal monoid monomorphisms as a certain kind of actions that we introduce here and illustrate with several examples.